PAPER_703: NGC 1275 (Perseus A): Magnetic Monster Black Hole Feedback UQFF

Class: `NGC1275MagneticMonsterUQFF` **CP4 Entry:** #287 **Keywords:** NGC 1275, Perseus A, Seyfert 1.5, magnetic filaments, BH feedback, MUGE, UQFF **Session:** 175 | **Version:** v5.32 **Source:** grok_share_ba508f76c8e.txt

Abstract

NGC 1275 (Perseus A), also called the Magnetic Monster, is a Type 1.5 Seyfert galaxy in the Perseus Cluster at $z = 0.0176$ (237 Mly). It hosts a supermassive black hole of $8 \times 10^8 M_\odot$ and exhibits spectacular Hα filaments extending 50 kpc, sustained by magnetic fields $B_{fil} \approx 10^{-8}$ T against thermal convection. The UQFF MUGE is derived incorporating BH feedback suppression and filament support.

1. System Parameters

Parameter	Value	Description
M_{SMBH}	$8 \times 10^8 M_\odot$	Supermassive black hole
$M_{stellar}$	$10^{12} M_\odot$	Stellar mass
r_{gal}	9.46×10^{20} m	Galaxy radius
z	0.0176	Redshift
B_{fil}	10^{-8} T	Filament magnetic field
τ_{BH}	3.156×10^{15} s	BH feedback timescale (~ 100 Myr)

2. Master UQFF Gravity Equation

$$g_{NGC1275}(r,t) = \frac{G(M_{SMBH} + M_\star)}{r^2}(1+H_z t)(1-F_{BH}(t))(1+f_{TRZ})+a_{fil}+q(v\times B)\left(1 + \frac{\rho_{UA}}{\rho_{SCm}}\right)$$

2.1 Black Hole Feedback Function

$$F_{BH}(t) = 0.1\left(1 - e^{-t/\tau_{BH}}\right)$$

This models the AGN-driven mechanical feedback that suppresses star formation.

2.2 Hubble Parameter at $z = 0.0176$

$$H(z) = H_0\sqrt{0.3(1+z)^3 + 0.7}$$

2.3 Filament Magnetic Support

$$a_{fil} = \frac{B_{fil}^2}{2\mu_0}\frac{V_{fil}}{M_{fil}} \times 10^{-12} \approx 2.840 \times 10^{-9}\ \text{m/s}^2$$

3. Numerical Result

$$g_{NGC1275}(t \approx 50\ \text{Myr}) \approx 3.16 \times 10^{-5}\ \text{m/s}^2$$

4. Physical Significance

The Hα filaments in NGC 1275 are direct observational evidence for magnetic field support against gravitational collapse and thermal convection. The UQFF framework naturally incorporates this via the a_{fil} term, with the vacuum aether (ρ_{UA}, ρ_{SCm}) providing the quantum substrate for magnetic tension.

References

- Fabian et al. (2008), Nature, 454, 968 (filament discovery)
- Churazov et al. (2000), A&A, 356, 788 (Perseus cooling flow)
- UQFF Framework, Star-Magic Session 175

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